

Slide 2: Design-Dependent Mission Strategy

Context: The selected rover design directly changes how the mission handles routing preference, power usage, thermal survival, and science reach.

Mission Layer	Solar-Optimized Rover Strategy	RTG-Based Rover Strategy
Routing Priority	Prefer safe paths with better solar illumination exposure	Prioritize safe terrain and ice-rich targets without depending on illumination
Power Management	Battery-aware movement, recharge windows, and sunlight-based duty cycles	Continuous power budget with longer operational windows and reduced recharge constraints
Thermal Management	Use electrical heaters carefully and avoid long cold-soak periods in shadow	Use radioisotope decay heat as thermal support for electronics and instruments
Science Reach	Best suited for PSR boundaries and accessible illuminated regions	Better suited for deeper PSRs, cold traps, and extended ice exploration
Main Mission Risk	Energy shortage and thermal survival in shadowed terrain	Higher mass, cost, safety, and approval complexity

Design Implication: The solar-optimized rover needs illumination-aware planning. The RTG-based rover allows greater freedom in low-illumination terrain but increases mission complexity.

Key Takeaway: The mission will use one final rover design, selected after comparing feasibility, survival margin, and science reach.